**25901404** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

## Fourth SEMESTER

## Discipline Specific Elective (DSE) - MG4DSEMBG202 - DAIRY MICROBIOLOGY

(2024 ADMISSION ONWARDS)

**Duration: 2 hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** marks

1. Name two microorganisms used as starter cultures in dairy microbiology. [K] / [CO1]
2. What is shelf life? [K] / [CO2]
3. What is the role of rennet in cheese production? [K] / [CO3]
4. Mention the concentration of Formalin in composite milk sample as effective antiseptic and preservative. [K] / [CO4]
5. Define standardized milk. [K] / [CO2]

**[1x5 = 5]**

### Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

6. Comment on the sources of contamination of milk. [K] / [CO1]

7. Write two sources of microbial contamination in dairy plants. [K] / [CO2]

8. Name the uses of whipped cream in food preparation. [K] / [CO3]

9. Define Pasteurized milk. [U] / [CO4]

10. Extend MBRT. [U] / [CO4]

11. What are thermoduric bacteria, and what is their significance in pasteurized milk? [K] / [CO1]

12. Mention any two microbial risks associated with milk distribution [K] / [CO2]

13. Which microorganisms are mainly responsible for souring of milk? [K] / [CO3]

14. Mention two objectives of milk drying. [K] / [CO2]

15. What are kefir grains composed of? [K] / [CO3]

16. Comment on HTST. [U] / [CO1]

17. Write two objectives of cream separation. [K] / [CO2]

**[2x10 = 20]**

## Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

18. Describe the characteristics of Clostridium species associated with milk and dairy products. [U] / [CO3]

19. Describe the production process of evaporated milk. [K] / [CO2]

20. What is homogenisation of milk?What is the difference between single-stage and two-stage homogenization? [U] / [CO1]

21. Explain in detail the types of spoilage of milk. [K] / [CO1]

22. Describe the role of lactic acid bacteria in milk fermentation. [U] / [CO3]

23. Explain the characters and preparation process of acidophilus milk. [U] / [CO3]

24. Write in detail about Agar Slice Methods for milk processing equipment. [U] / [CO4]

**[6x5 = 30]**

## Part D

Essay Questions

Answer any **1** questions

Each question carries **15** marks

25. Explain the scope and importance of Dairy Microbiology. [U] / [CO1]

26. Explain the characteristic properties of milk with reference to its composition and properties. [K] / [CO1]

**[15x1 = 15]**